DATABASE SCHEMA DOCUMENTATION

Mini Client Management System

1. Database Overview

The Mini Client Management System uses MongoDB as its database and Mongoose as the Object Data Modeling (ODM) library.

MongoDB stores administrator authentication details and client information in separate collections.

Database Details:

- Database: MongoDB
- ODM: Mongoose
- Hosting: MongoDB Atlas
- Database Type: NoSQL
- Collections: Users and Clients

2. Database Collections

The application contains two main collections:

1. Users Collection
2. Clients Collection

3. Users Collection Schema

The Users collection stores administrator account information required for authentication.

Field Name	Data Type	Required	Description
_id	ObjectId	Auto	Unique user identifier
name	String	Yes	Administrator name
email	String	Yes	Administrator email address
password	String	Yes	Administrator password
role	String	No	User role (Default: admin)
createdAt	Date	Auto	Account creation date
updatedAt	Date	Auto	Last updated date

Validation Rules:

- Name is mandatory.
- Email is mandatory and must be unique.
- Email is automatically converted to lowercase.

- Name and email are trimmed to remove extra spaces.
- Password is mandatory.
- Default user role is admin.
- Timestamps are automatically generated.

4. Clients Collection Schema

The Clients collection stores information about registered clients.

Field Name	Data Type	Required	Description
_id	ObjectId	Auto	Unique client identifier
name	String	Yes	Client name
email	String	Yes	Client email address
phone	String	Yes	Client phone number
company	String	Yes	Company name
status	String	No	Active or Inactive
address	String	No	Client address
notes	String	No	Additional information
createdAt	Date	Auto	Record creation date
updatedAt	Date	Auto	Last updated date

Validation Rules:

- Client name is mandatory.
- Email is mandatory and must be unique.
- Email is automatically converted to lowercase.
- Phone number is mandatory.
- Company name is mandatory.
- Status accepts only Active or Inactive.
- Default status is Active.
- Address and notes are optional.
- Timestamps are automatically generated.

5. Sample User Document

{

"_id": "ObjectId",

```
  "name": "Admin",
  "email": "admin@example.com",
  "password": "hashed_password",
  "role": "admin",
  "createdAt": "2026-11-04T10:00:00.000Z",
  "updatedAt": "2026-11-04T10:00:00.000Z"
}
```


6. Sample Client Document


```
{
  "_id": "ObjectId",
  "name": "Arun Kumar",
  "email": "arun@example.com",
  "phone": "9876543210",
  "company": "ABC Technologies",
  "status": "Active",
  "address": "Puducherry, India",
  "notes": "Regular client",
  "createdAt": "2026-11-04T10:00:00.000Z",
  "updatedAt": "2026-11-04T10:00:00.000Z"
}
```


Note: The sample documents above are illustrative examples. The password should be securely hashed before storage.

7. Database Operations

The application performs the following database operations:

Users Collection:

- Create administrator account.
- Retrieve administrator details.
- Validate login credentials.

Clients Collection:

- Add new client records.
- Retrieve all registered clients.

- Search clients by name, email, or company.
- Update existing client information.
- Delete client records.
- Filter clients based on Active or Inactive status.

8. Conclusion

The MongoDB database provides an efficient way to store and manage administrator and client information. Mongoose schemas ensure data validation, consistency, and structured database operations throughout the Mini Client Management System.